

Web Design

Session 4 – Components, Icons & Images

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Warm-up & Recap

Create **clear, ordered** and **resilient**
designs with alignment and
simple systems.

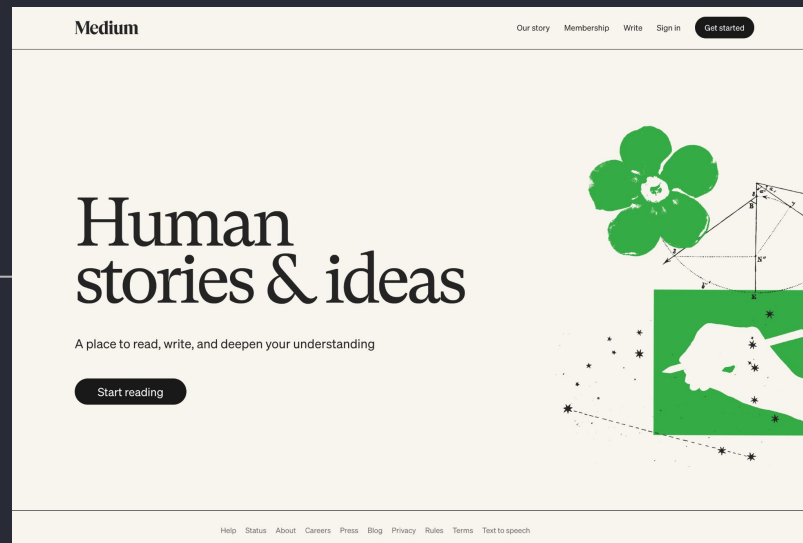
What makes good web layouts?

Hierarchy through

Spacing & Alignment

Structure with

Predictability vs. Creativity



Good layout = **clarity** + **visual rhythm**

BUY JAMIE'S NEW COOKBOOK

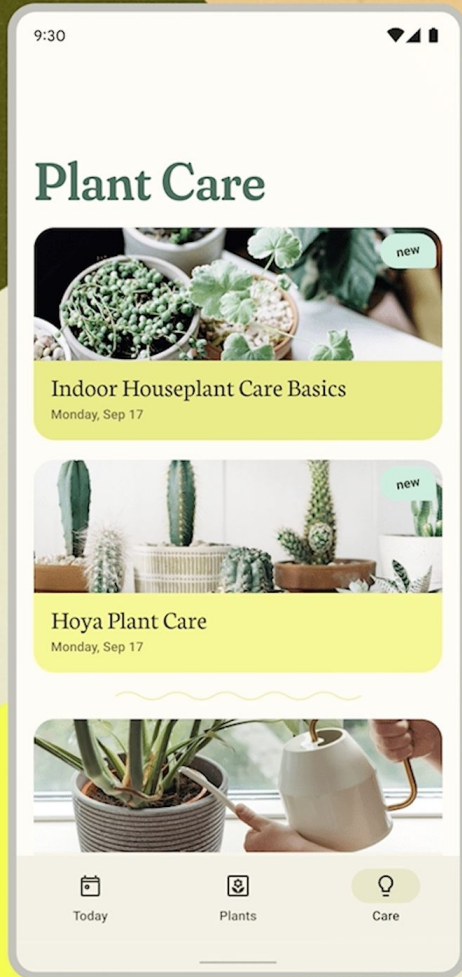
Eat Yourself Healthy will arm you with ideas and inspiration that will energise, satisfy and revitalise you

BUY NOW



Summary

- Use same spacings → consistency
- Use grids
- Design mobile first
- Atoms → Components → Design System



What was your biggest learning until now?



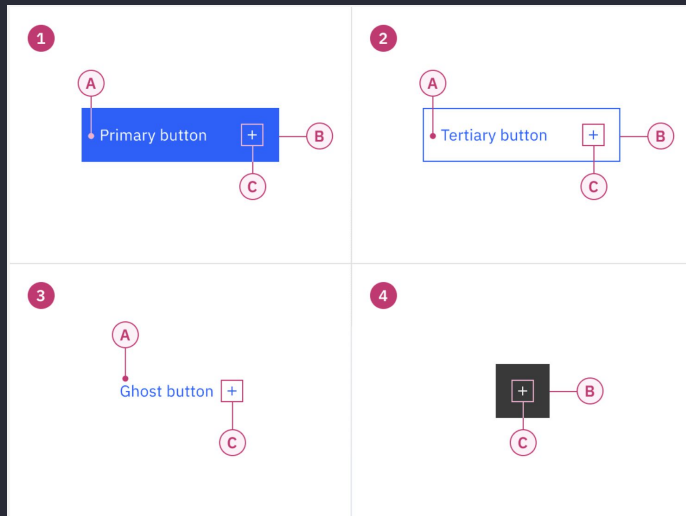
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Session 4

Components, Icons & Images

- Components, variants, reusability in design
- Component vs. instance
- Best practices for images and SVGs
- Using component / icon libraries
- Preparing assets for development
- Accessibility overview



The Goals

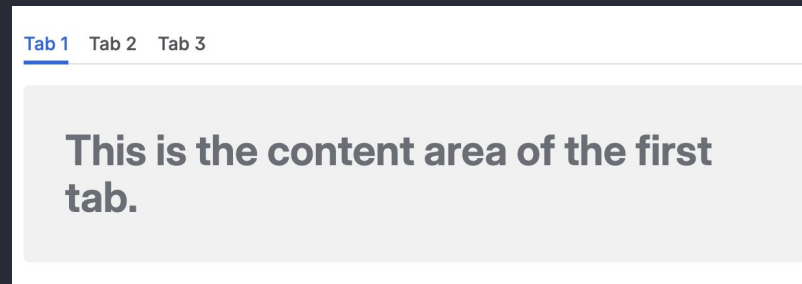
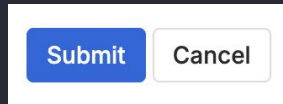
Understand **reusable** design structures

Manage **assets** (icons, images, SVG)

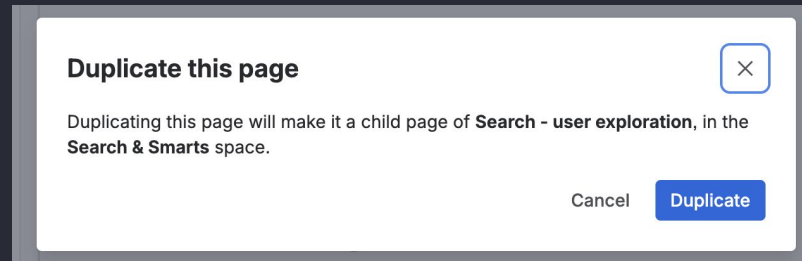
What is a *component*?

What is a component?

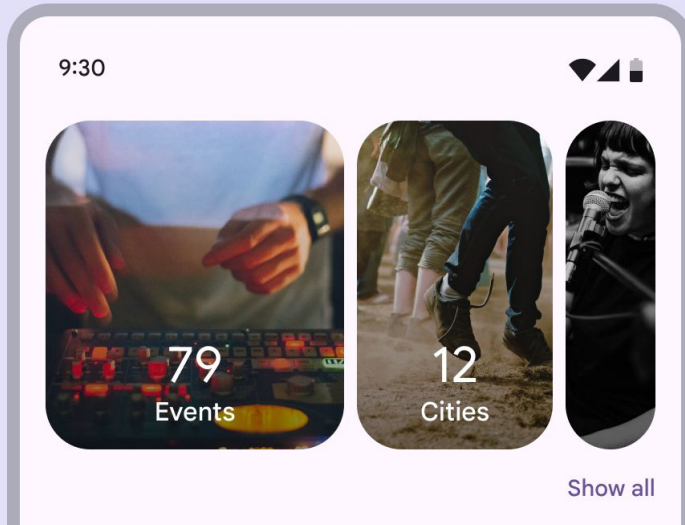
- A reusable element (e.g., button, card, navbar)
- Keeps design consistent
- Editable in one place



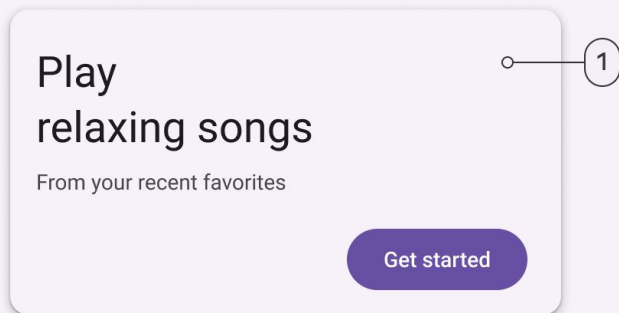
✓ **Merged pull request**
Pull request #10146 merged after a successful build
[View commit](#) · [Dismiss](#)



Carousel

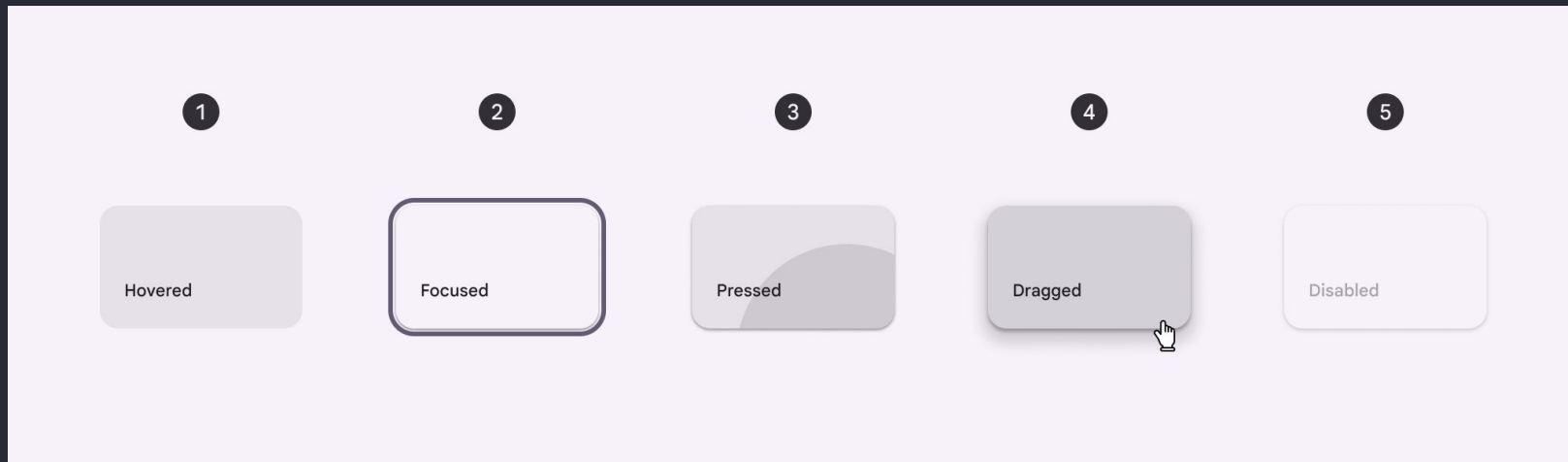


Card



Variants of a component

- A component often has different states
e.g. default / hover / disabled / dark mode



Component vs. instance

- Component = “Master Plan”
 - defines look and behavior
 - defines properties which can be different in each instance
 - e.g. “VW Golf 8” (**motor**, **color**)
- Instances = content-filled representations
 - override properties (Texts, Images, ...)
 - e.g. “VW Golf **1.4 TSI**, **white**” or “VW Golf **2.0 TSI**, **red**”

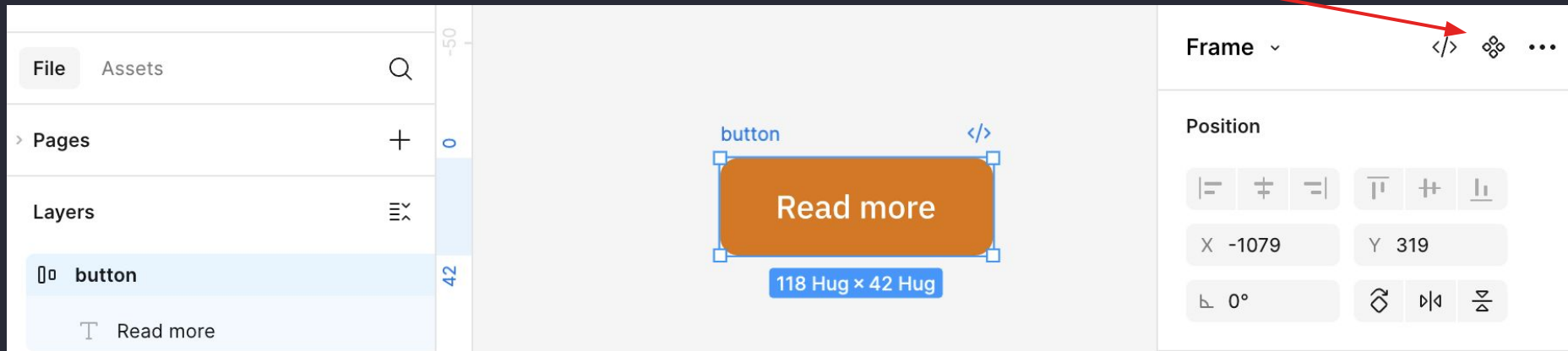
Variants of a button component

	Default	Default	Focus	Loading
Primary	Button label	Button label	Button label	Button label 
Primary + Icon Left	+ Button label	+ Button label		
Primary + Icon Right	Button label +	Button label +		
Secondary	Button label	Button label		Button label 
Secondary + Icon Left	+ Button label	+ Button label		
Secondary + Icon Right	Button label +	Button label +		
Icon Only				

Components in Figma

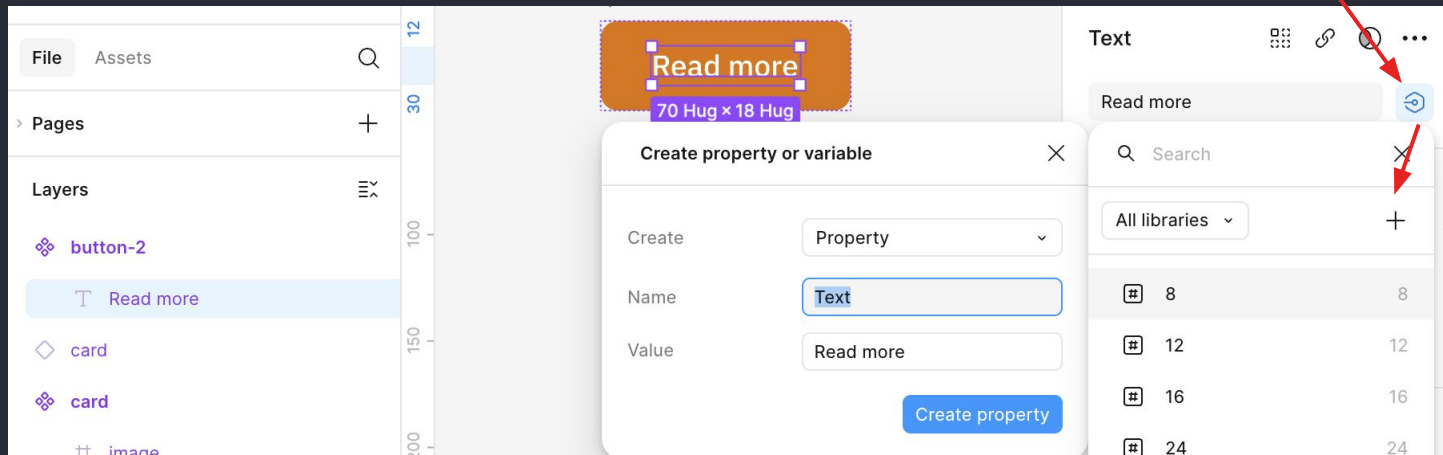
Create a component

- Select group of layers (in that case an Auto-Layout)
- "Create component" - CMD + ALT + K



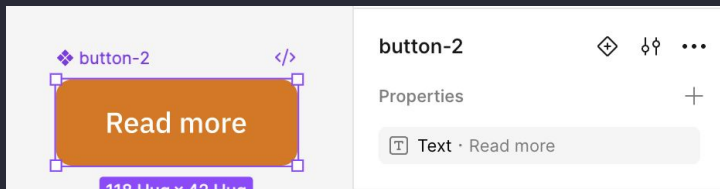
Add properties

- Select text layer → “Apply layer/property” → +
- Name your property
- This will be available to override in instances!



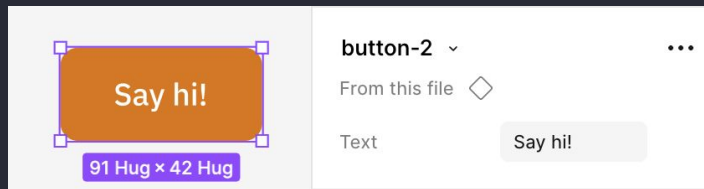
What we've created

- A component "button"
- with one property "text" that can be overridden in button instances



Component

- filled diamond icon

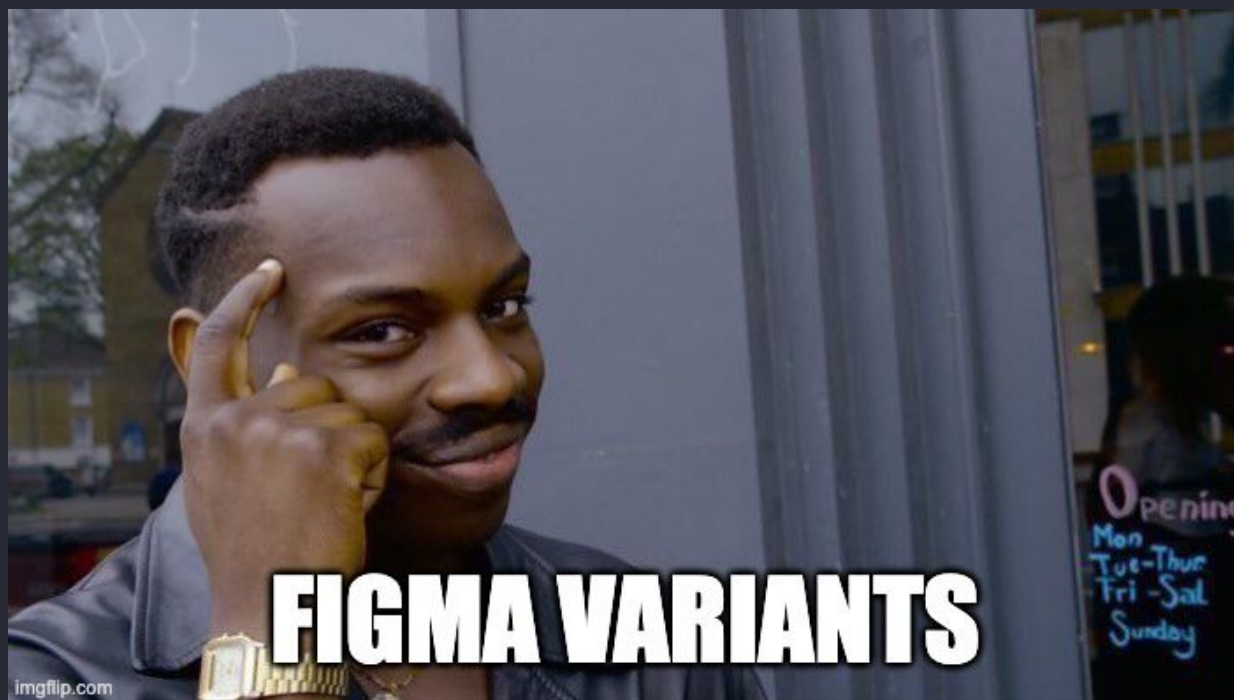


Instance

- hollow diamond icon
- customized text property

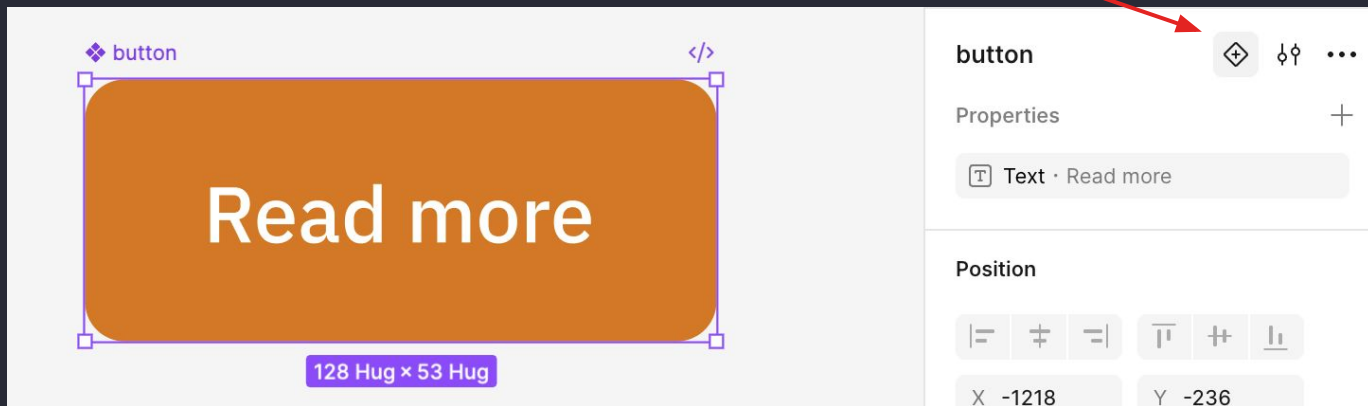
WHAT IF

YOU NEED A BOATLOAD OF BUTTONS

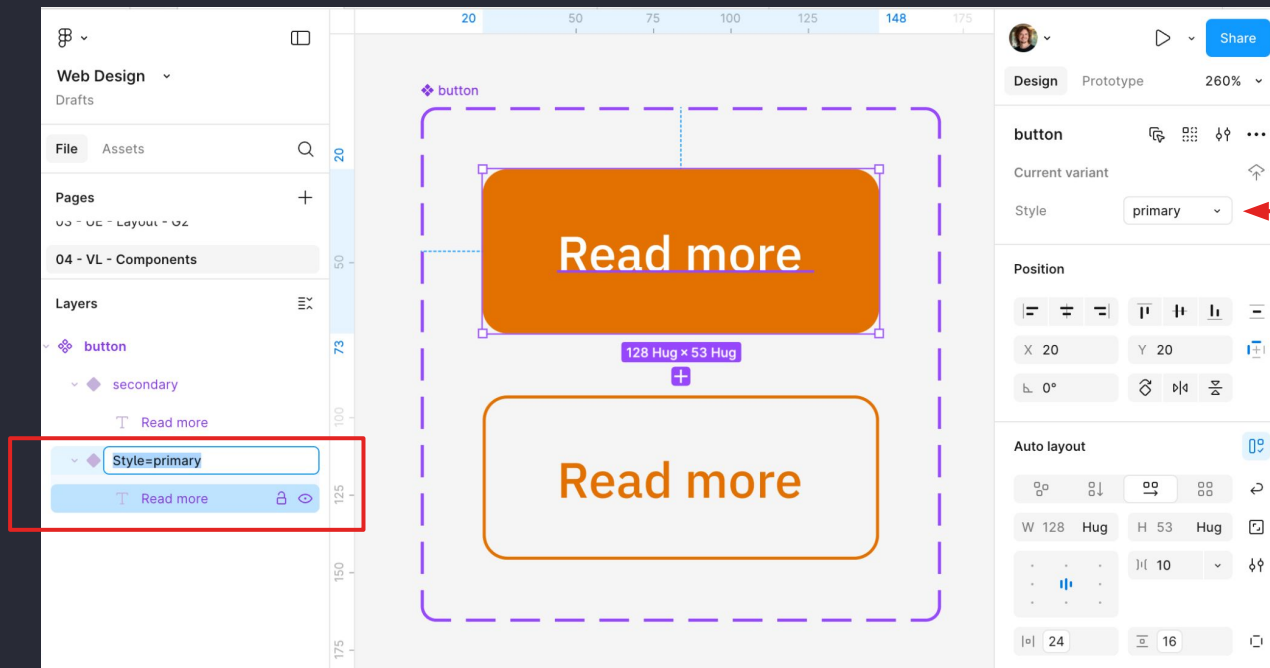


Variants in Figma - Create

- Variants can grow fast, keep it simple
- Create component → “Add variant”



Variants in Figma - Name

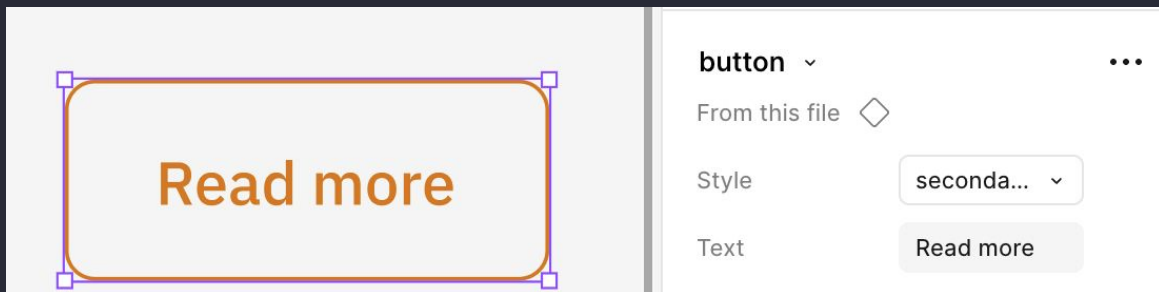


Rename "Property1" to "Style"

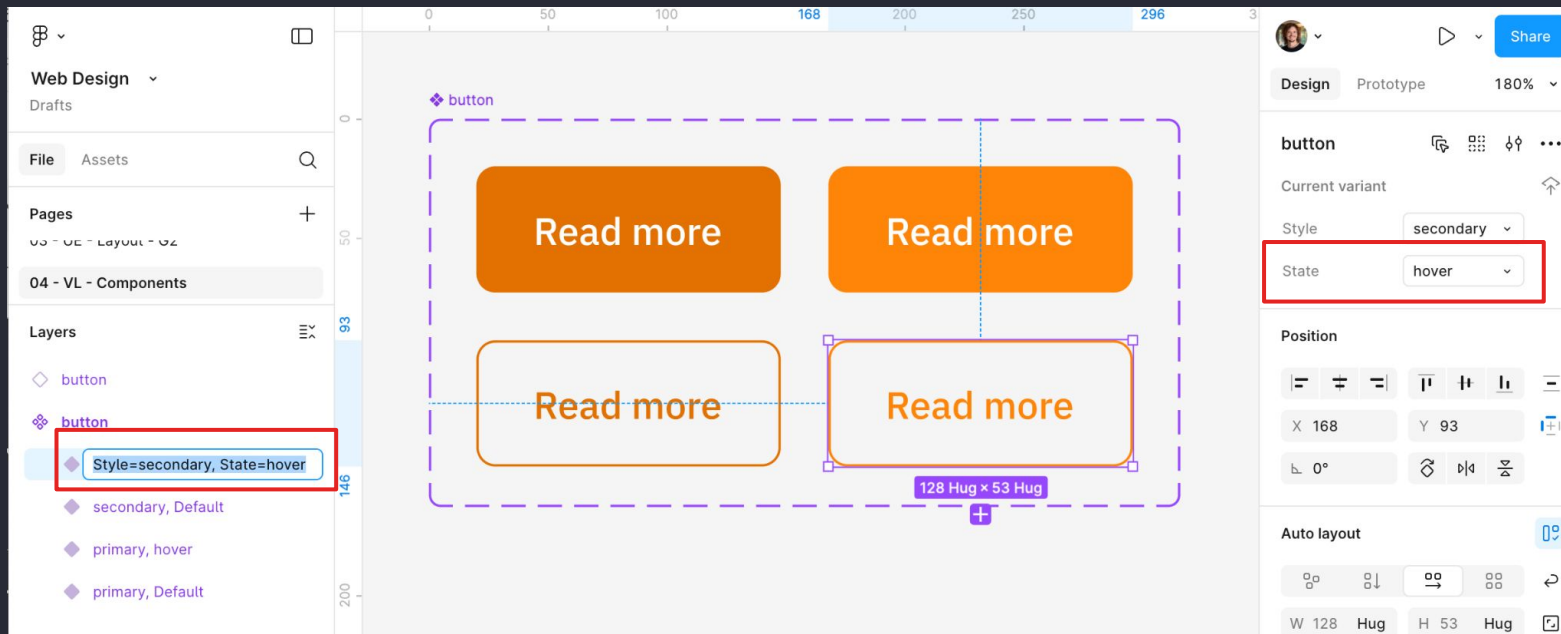
Rename "Default" to "primary"

Variants in Figma - Use

- Shift + I → Search “button” → Insert → Choose “Style - secondary”

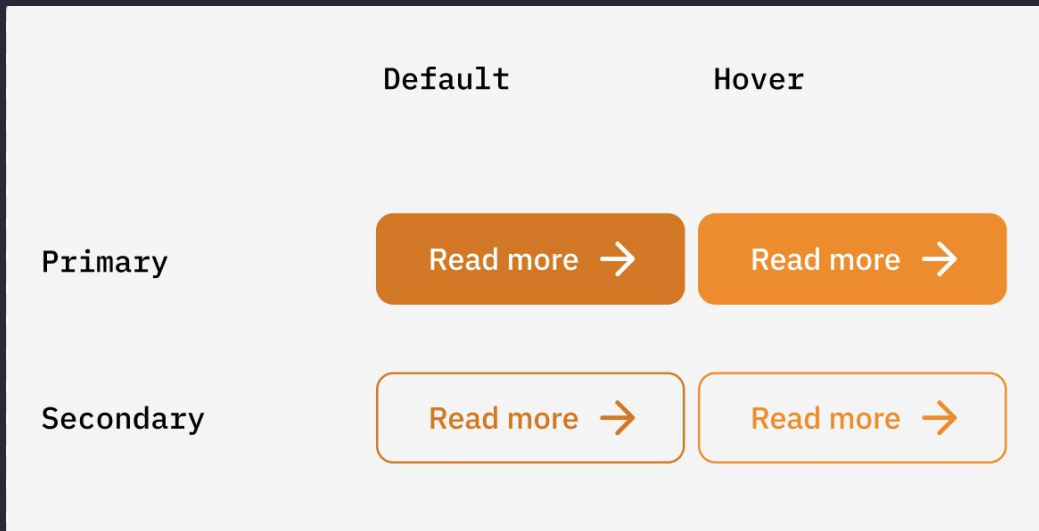


Variants in Figma - Multiple Variants



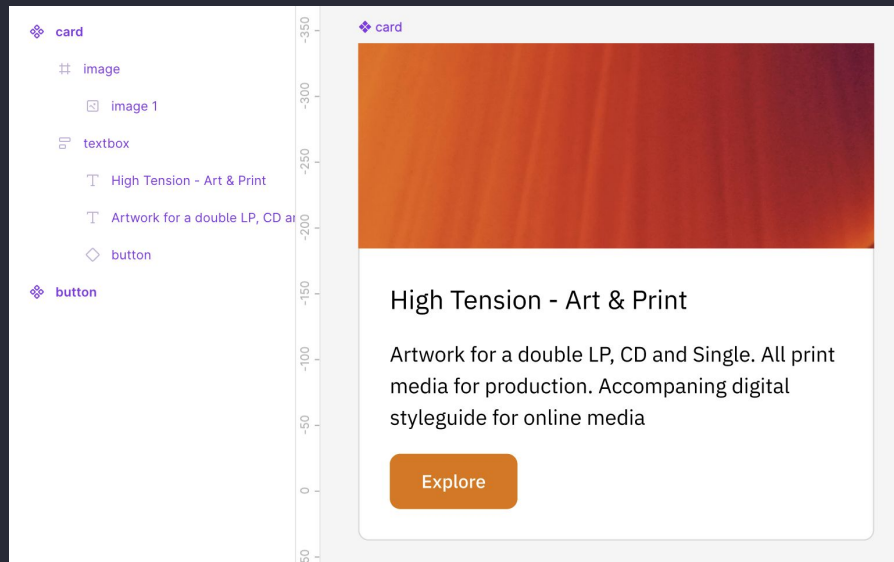
Component and Variants

- Property Text
- Property Icon
- Variant Style
 - o primary
 - o secondary
- Variant State
 - o *default*
 - o *hover*



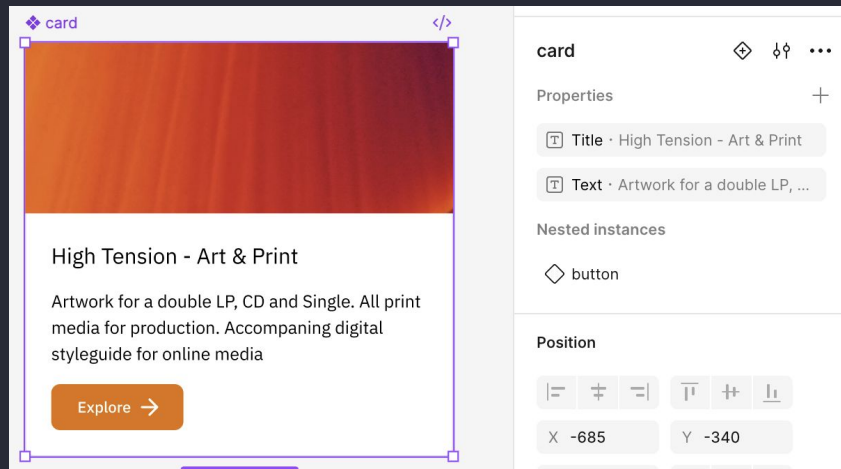
Nesting components

- Combining small components → larger components
- Keep consistency
- “button” inside “card”



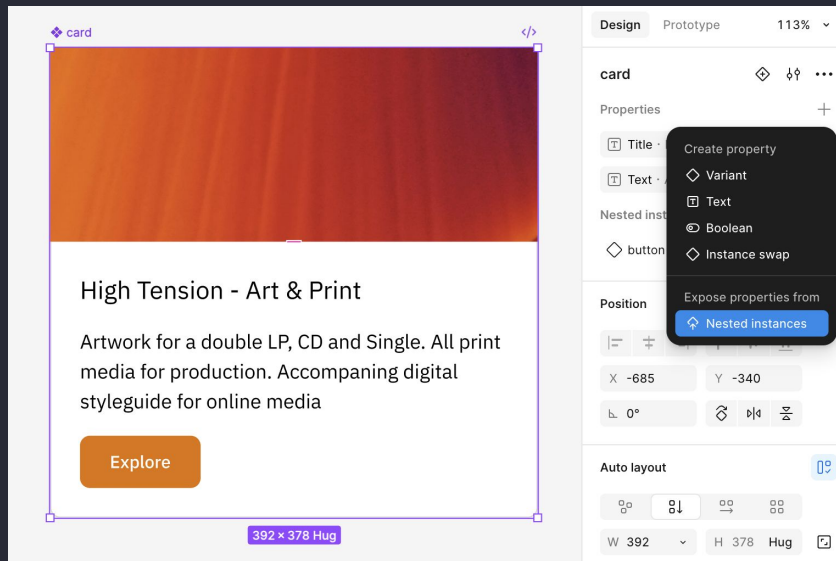
Nesting components

- Only *Title* and *Text* from card
- You want to override the “Explore” of the *button*

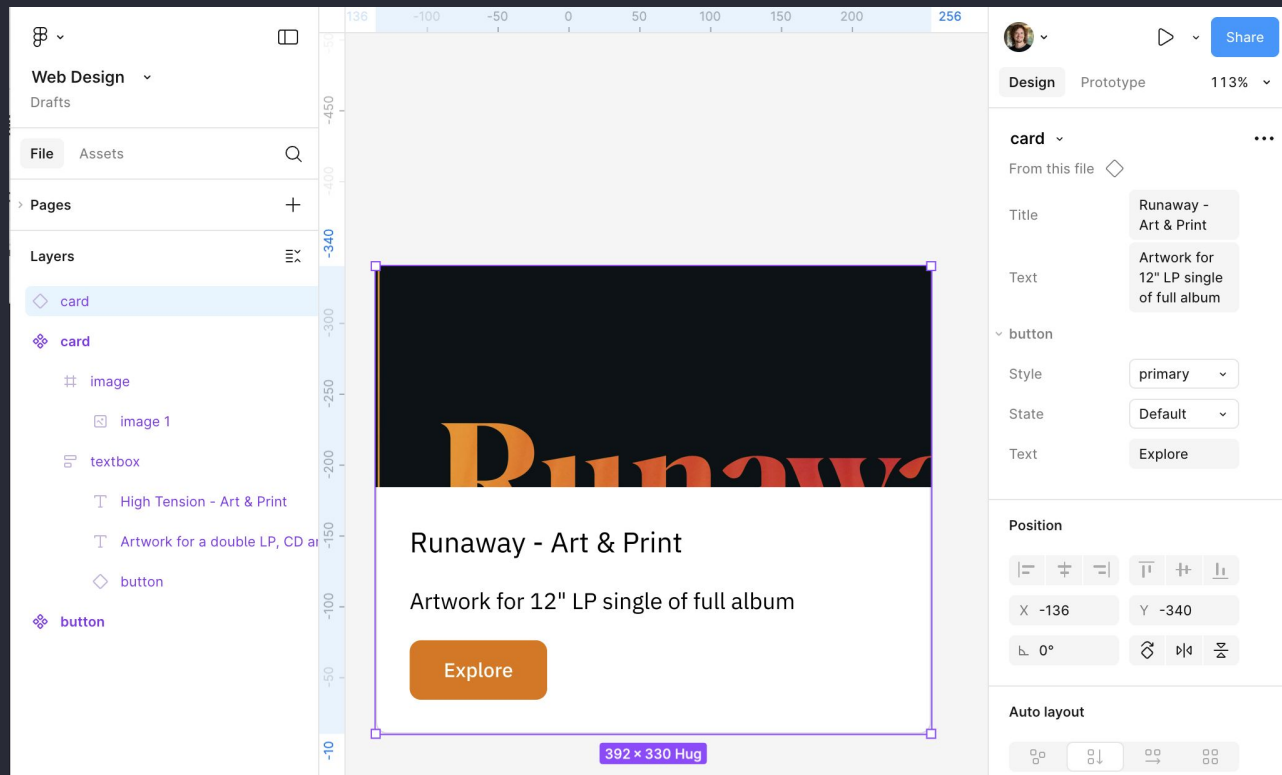


Nested instance properties

- Expose the button's properties to the card
- Select the card, Add new property → "Nested instances"



Nested instance properties



Each card instance can have a different button

Summary – Components

- Component is a reusable piece of design
- A component has one or more **properties**
 - e.g. *Text, Description*
- One component, multiple **instances**
- **Variants** for different states of a component
 - e.g. *Style (primary, secondary), State (default, hover, focused)*

Components in HTML

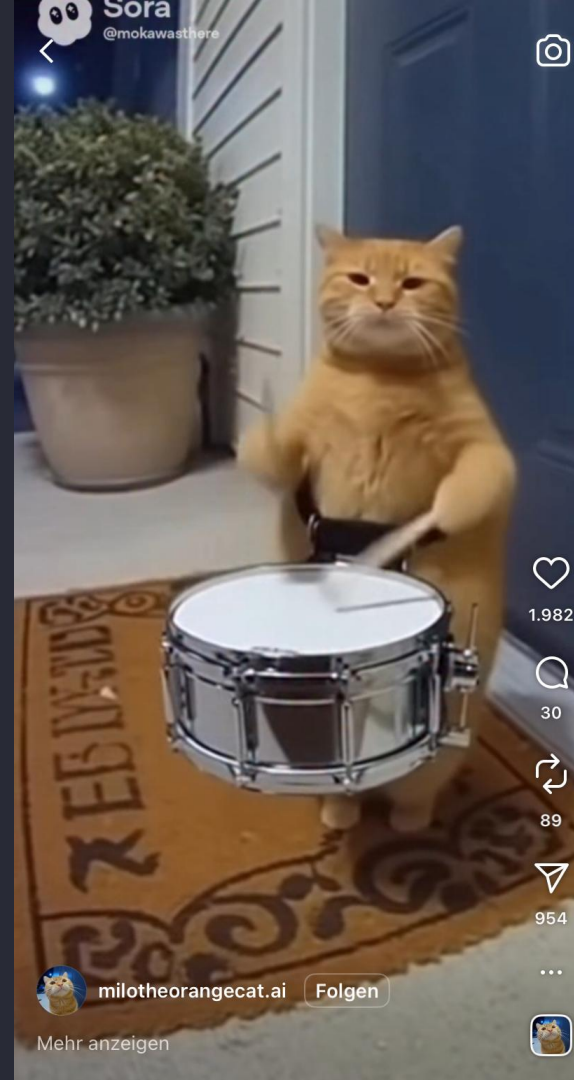
```
<a href="https://google.com" target="_blank">Google</a>
```

Properties 🥰 actually “data attributes” but same principle!

Components in Javascript

```
function hello() {  
  const cat = {  
    name: 'Milo',  
    age: 1,  
    meow: function() {  
      console.log(`Meow! My name is ${this.name}`);  
    }  
  };  
  
  cat.meow();  
}  
  
hello();
```

Properties 🥰



**Where do you see
the biggest benefit
of using components?**

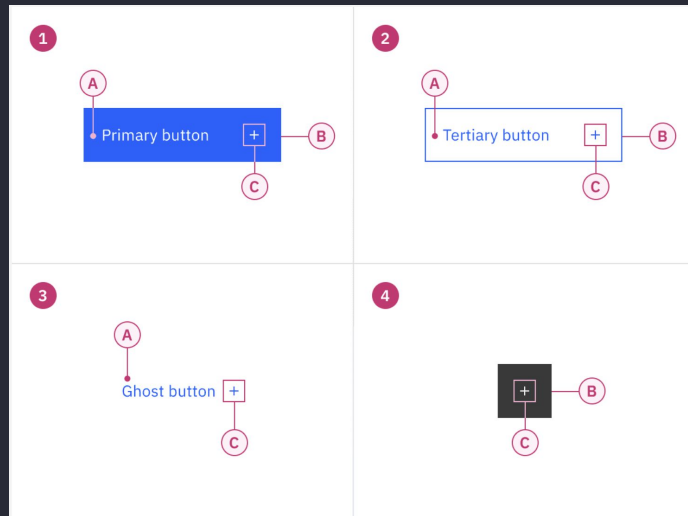


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Components, Icons & Images

-  Components, variants, reusability in design
-  Component vs. instance
- Best practices for images and SVGs
- Using component / icon libraries
- Preparing assets for development
- Accessibility overview



Best practices for Images & SVGs



LOST PUPPY - PLEASE HELP!
URGENT
Our dog "Buddy" went missing
here. He is friendly but scared.
If seen. If seen, call this number
IMEDIATELY: 555-PET-FIND.

Image Formats

JPG

SVG

PNG

WebP

Image Formats - JPG

- photos, large visuals
- raster format (scaling bigger → less quality)
- lossy compression = you lose information (blurry)
- no transparency
- small file size

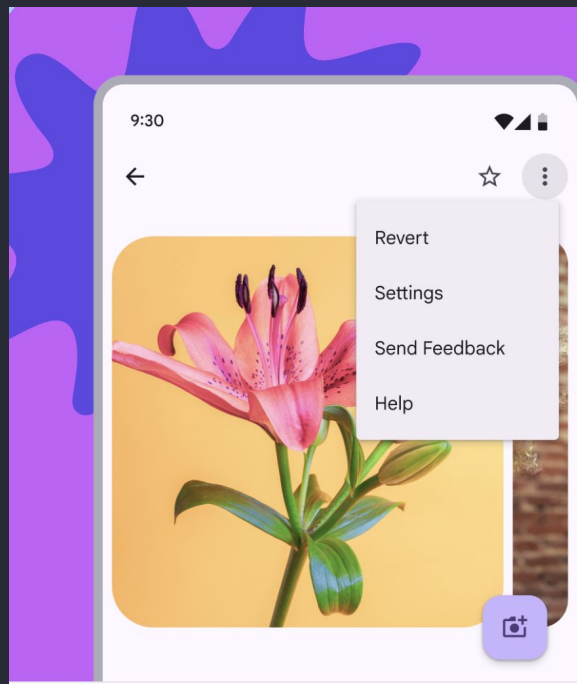


Image Formats - PNG

- transparent UI elements , small to midsize images
- raster format
- lossless compression (sharp, no blur)
- transparency
- larger file size

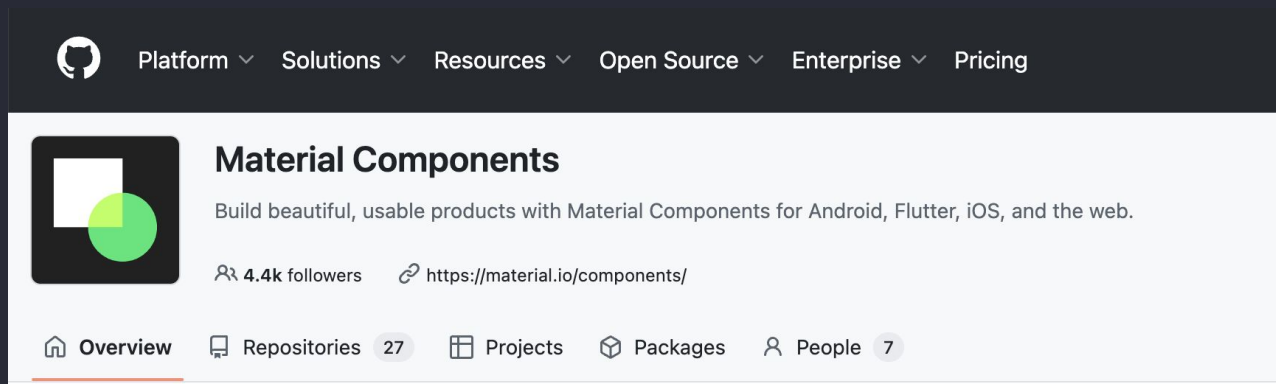


Image Formats – SVG

- logos, icons, illustrations
- don't use for “photos”
- vector format (lossless scaling)
- lossless compression (sharp, no blur)
- transparency
- small file size



material-components-android

Public

Modular and customizable Material Design UI components for Android



Java



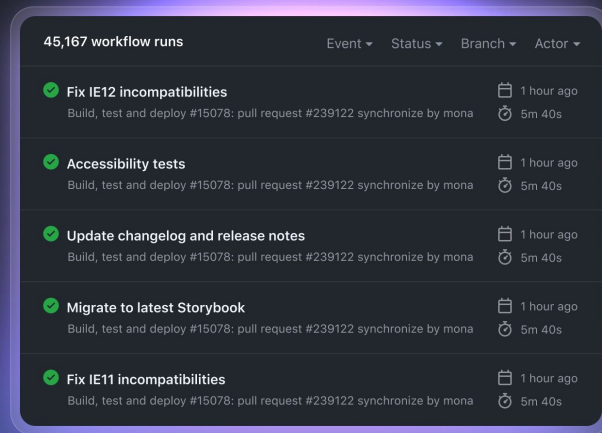
17k



3.2k

Image Formats - WebP

- optimized web format, *prefer over JPG and PNG*
- raster format
- lossy and lossless compression (region based)
- small file size

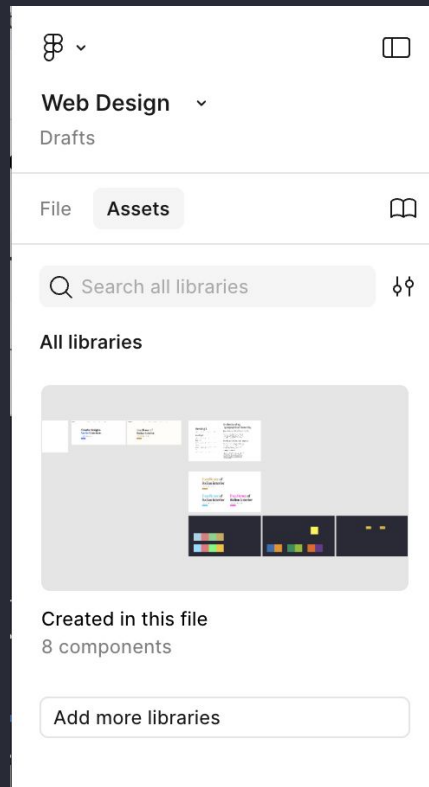


Resolution & Optimization

- You are exporting assets for development
- Talk to devs about sizes
 - Sizes in the implementation might differ from static designs
- Avoid oversized assets
- Compress with SVGO or TinyPNG, oftentimes development workflows are creating .webp from .jpg or .png sources
- <https://svgomg.net/>

Exporting images for development

- Consistent naming (icon_home.svg or hero-bg.jpg)
 - kebab-case or snake_case
- SVG for icons
- PNG for transparent images
- JPG for photos / larger images
- Organize in Figma “Assets” Panel

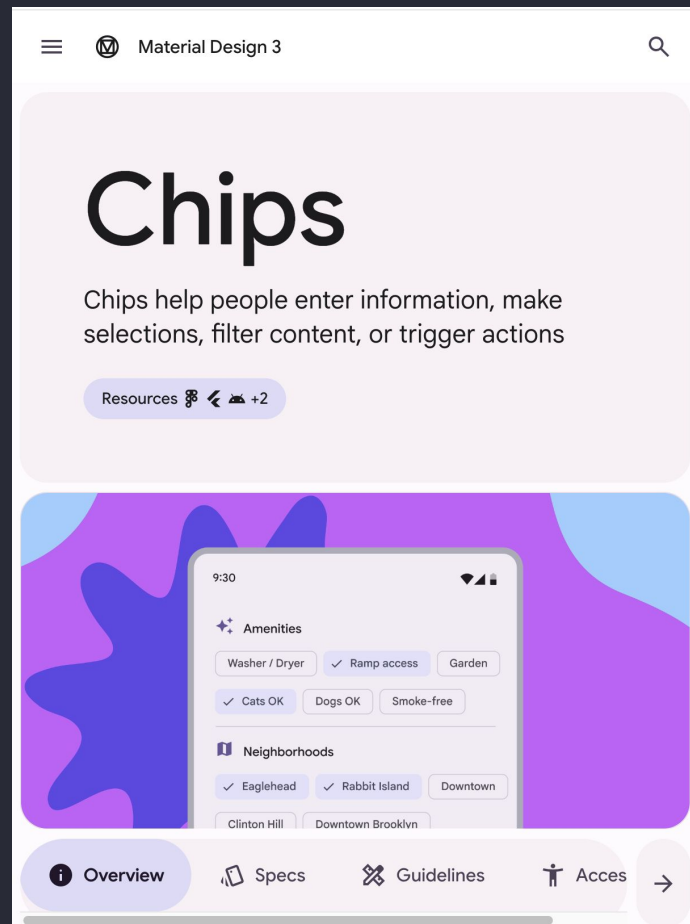
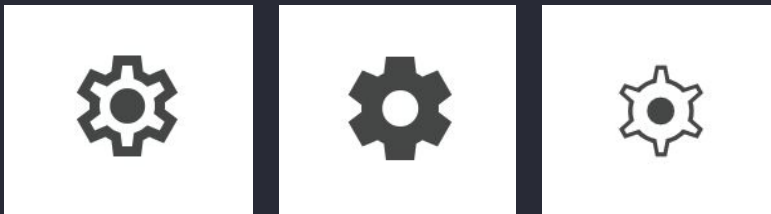


A detailed illustration of a futuristic library. The room is filled with tall, dark wooden bookshelves that stretch into the distance. In the center of the room is a large, round, dark wooden table. On the table, there are several open books, and a glowing, circular digital interface is projected onto its surface. The floor and the air around the table are filled with numerous floating, glowing icons of various shapes and colors, including Wi-Fi symbols, speech bubbles, and geometric shapes. Two figures in long, dark robes are standing near the table, interacting with the digital interface. The overall atmosphere is one of advanced technology and knowledge. The text "Using Icon Libraries" is centered in the middle of the image in a white, serif font.

Using Icon Libraries

Why icons?

- Support recognition and hierarchy
- Very important in Application UI
- Icons have a character similar to typography
 - o stroke vs. filled
 - o rounded vs. sharp corners
 - o visual weight



Why icons?

- Common web icon sets
 - <https://feathericons.com/>
 - <https://fonts.google.com/icons>
 - <https://lucide.dev/icons/>
- Prefer SVG over PNG
- Scalability and Accessibility

Customizer

Color  current

Stroke width 2px



Size 24px



Absolute Stroke width



Include external libs

☐ Lab

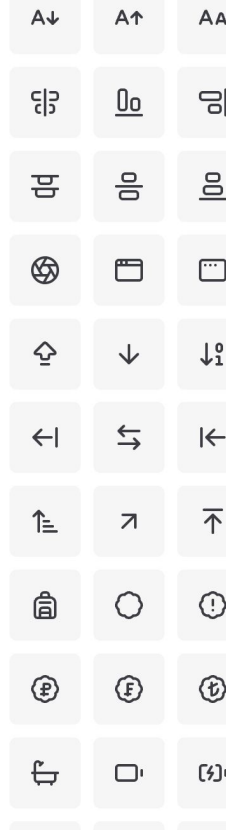
View

All

Categories

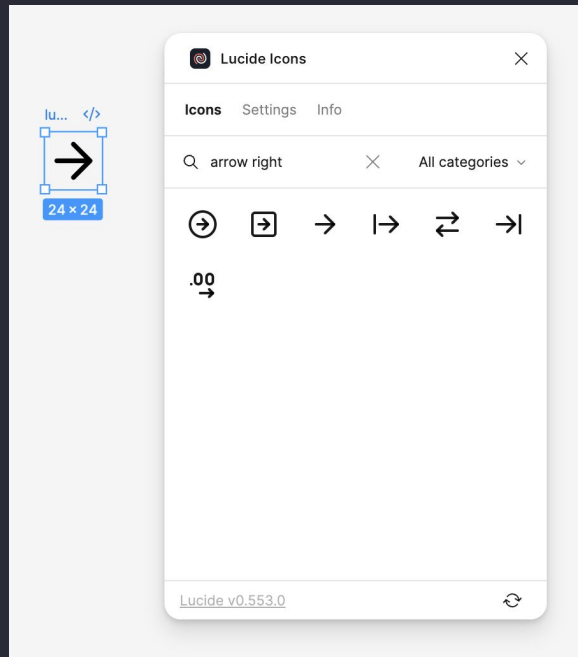
Accessibility	30
Accounts & access	133
Animals	23
Arrows	207
Brands	21
Buildings	25
Charts	31

 Search 1647 icons



Use icons via Figma Plugin

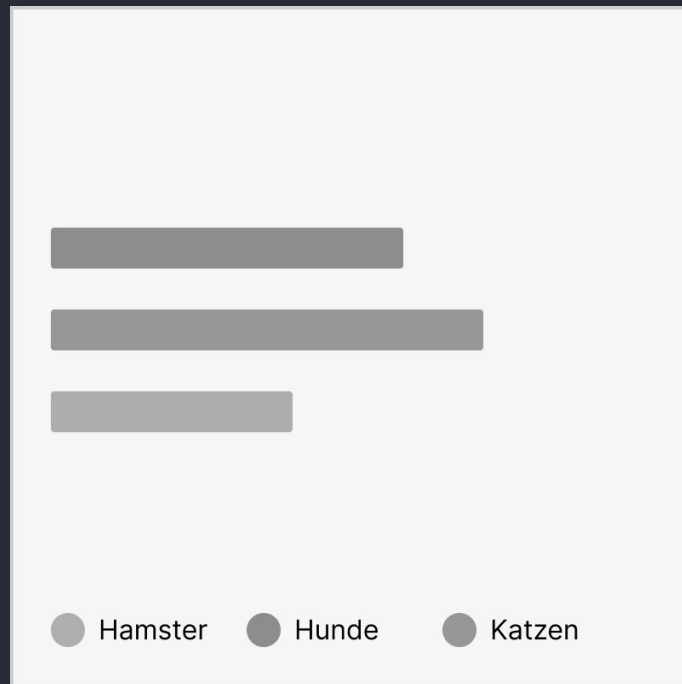
- Figma > Plugins > Manage Plugins > Search for “Lucide” > Run
- Plugins > Lucide
- Click on icon to insert



Accessibility in Visual Assets

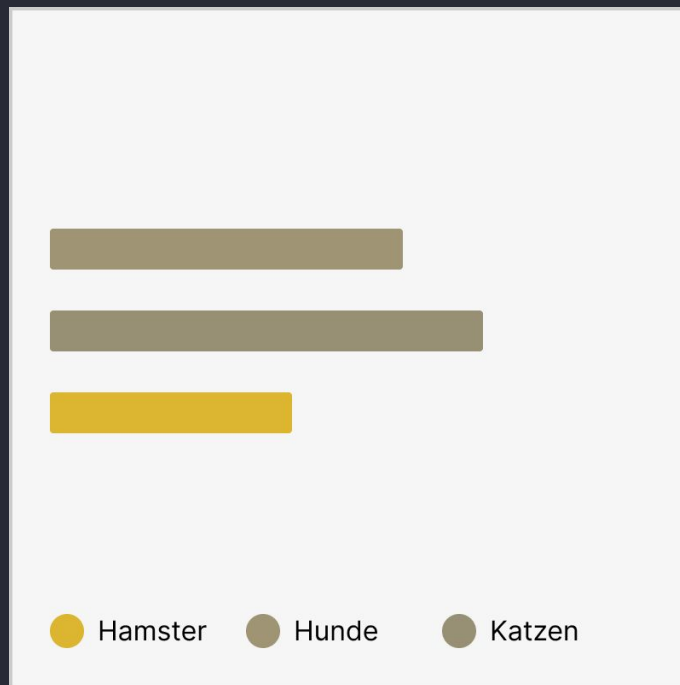
Attenzione

- Don't rely on color alone
- Images might not load
- Icons need a title or aria-label
- Add alternative texts to images
 - Purpose of the image!



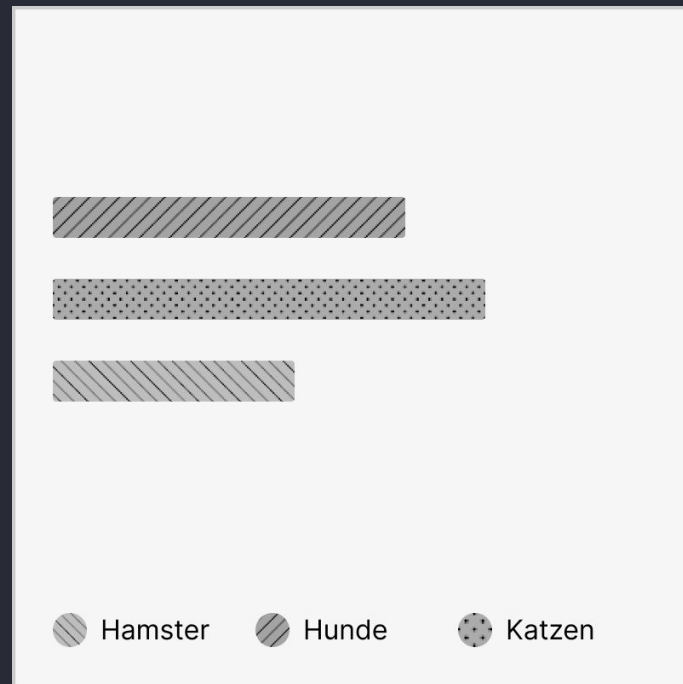
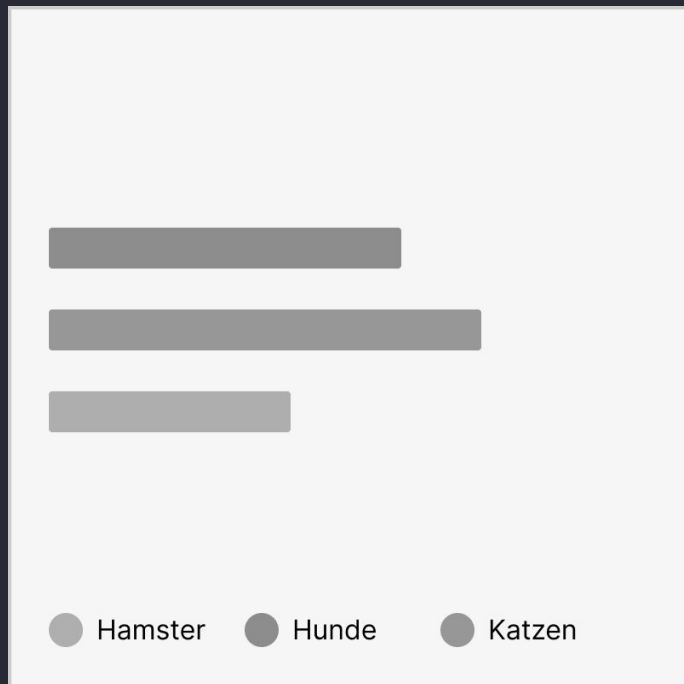
Simulation Achromatopsia (Color Blindness)

Don't rely on color alone



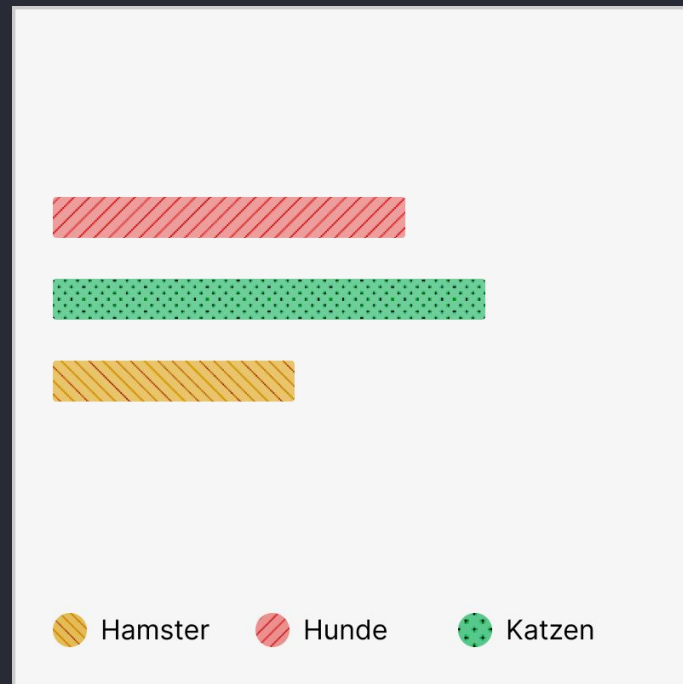
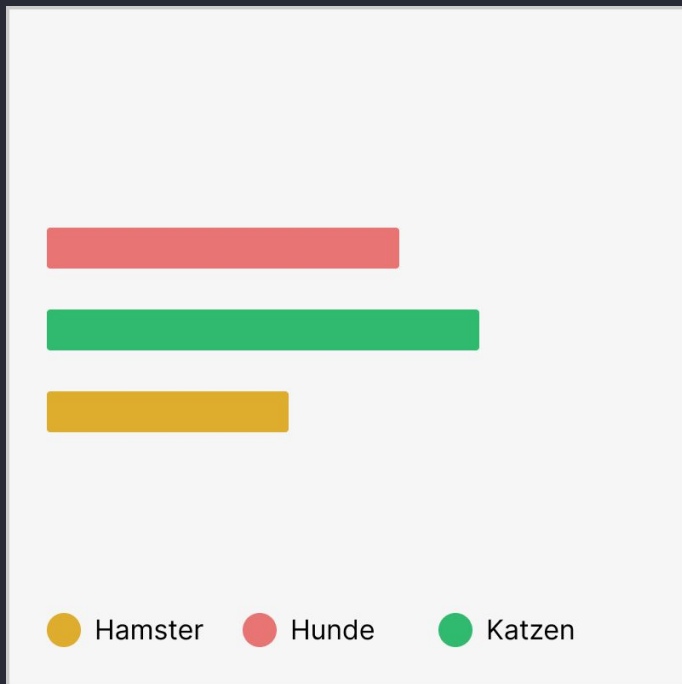
Simulation Deuteranopia (Form of red/green color weakness)

Don't rely on color alone

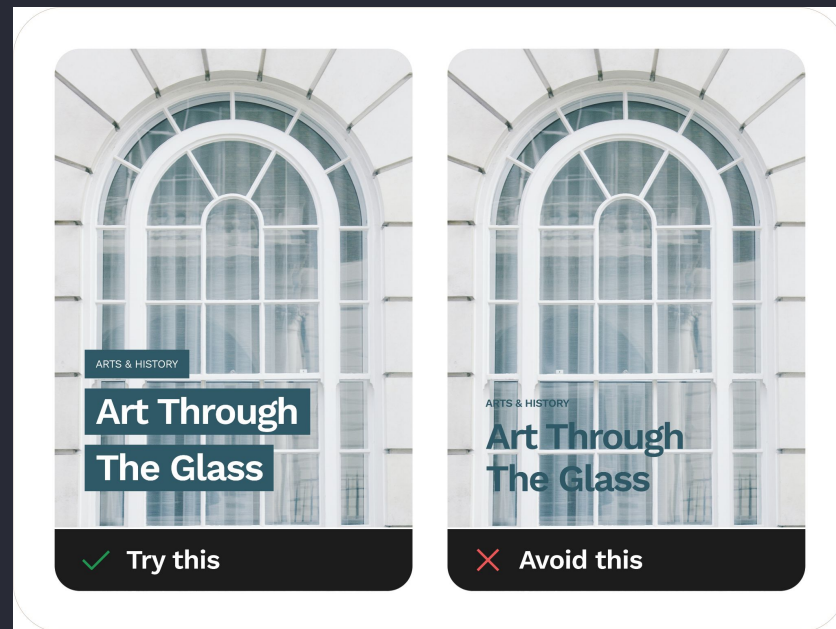
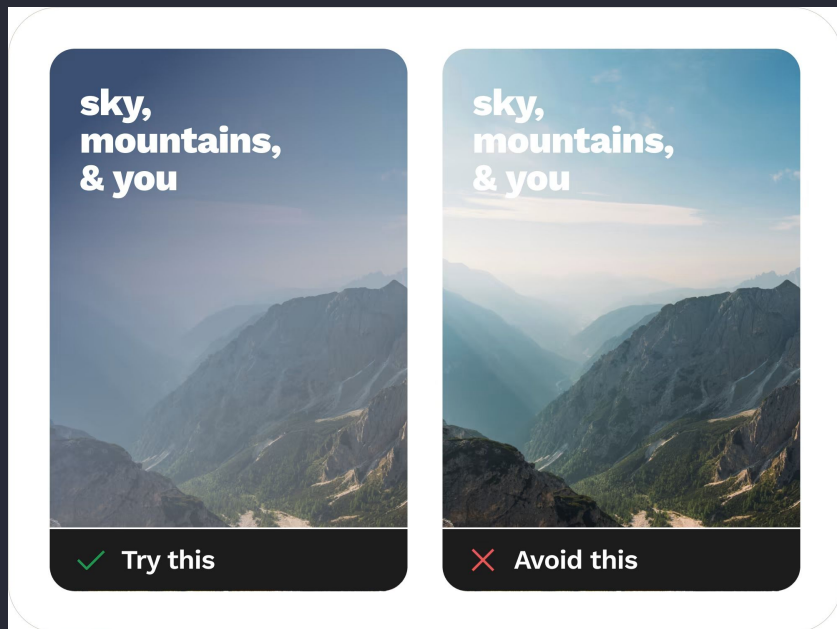


Simulation Achromatopsia (Color Blindness)

Don't rely on color alone



Ensure high contrast



Provide text alternatives to images and videos



What does this image show?

Provide text alternatives to images and videos



The stroke color was softened to improve visual hierarchy between chips and buttons.



Provide text alternatives to images and videos



SONOR. The Beat of Life. Our drums, orchestral and marching instruments are handcrafted in germany. A process that has been constantly developed and refined since 1875.

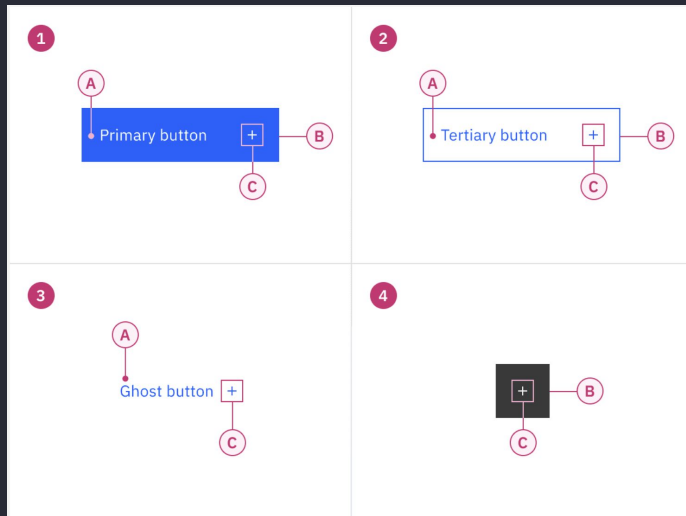
Animations

- Avoid flashing / dizzying effects
- Can irritate, trigger seizures
- Provide play / pause buttons

Wrap-Up

Components, Icons & Images

-  Components, variants, reusability in design
-  Component vs. instance
-  Best practices for images and SVGs
-  Using component / icon libraries
-  Preparing assets for development
-  Accessibility overview



Visual Design → Design Systems

Summary

- Understand reusable design structures
- Components, Instances, Variants
- Manage assets (icons, images, SVG)

Lab Session

- Create & use more components
- Use icons
- No hand-in this time

Homework Checklist				
	Wireframes		Design	
	Mobile	Desktop	Mobile	Desktop
Homepage	HW01		HW02	HW02
Project Detail Page			HW03	HW01
Navigation			HW03	HW03
Styleguide Colors & Typography	HW01			
WCAG AA Contrasts	HW01			
Apply Grids, Spacing and Auto Layout	HW02			
Component "Project Box"	HW02			
Component Button with Variants	HW03			
Component Navigation	HW03			
Component Header	HW03			
Component Footer	HW03			
Add Icons	HW03			
Build components for other repeated elements in your design				
Create a Prototype	HW04			
Create Handoff spec (Styleguide page)	HW04			
Export Assets	HW04			
Re-check Colors, Typography Styles, Grids, Spacings, A11y contrast, WCAG Easy Checks	HW04			

Thanks!